

SheetEX v2 — Teacher Guide

Data wrangling for AI across Excel 365, Excel 2013, Google Sheets, CSV/TSV and SQL.

8 lessons · 71 auto-checked challenges · about 10 class periods (50 min) · certificates for every lesson

Teacher-only: this guide contains the answer key. It is NOT part of the student app.

How it runs

- **Students** open your Apps Script web-app link (embedded in Canvas). Progress saves in their browser automatically.
- **Certificates** unlock when every challenge in a lesson is done. Students print/save as PDF, download an image, or paste the certificate code into Canvas.
- **Gradebook** (Apps Script only): every certificate is recorded in the *SheetEX Gradebook* Google Sheet in your Drive. Run `setup` once in the script editor to get its link.
- **Checking codes:** in the app click **? ► For teachers** and paste all submitted codes at once.
- **Hints** cost 20% of a challenge's XP; a certificate shows hints used, so you can see independence at a glance.
- **Moving computers:** students use *Class cloud save* (name + PIN) or copy their progress code.

Suggested pacing

#	Lesson	Challenges	XP	Time
1	Spreadsheet Foundations Excel 365	10	225	50 min
2	Classic Excel & Nested Functions Excel 2013	8	190	50 min
3	Google Sheets Survival Guide Google Sheets	9	205	50 min
4	Data Files: CSV & TSV CSV / TSV	8	135	45 min
5	Data Wrangling I: Cleaning Data for AI Data Wrangling Lab	11	220	90 min
6	Data Wrangling II: Combining & Lookups Data Wrangling Lab	11	220	90 min
7	SQL for Data Analysts SQL Database	9	215	75 min

8	Cross-Platform Translator Compare	5	60	25 min
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Lessons 1–4 can be done in any order. Lessons 5–6 (Data Wrangling) are the core of the AI-data unit — the  AI check tab shows students how their cleaning changes what a model learns. Lesson 8 makes a good review day.

Differentiation: early finishers can switch the Data Wrangling Lab engine (Excel 365 ⇄ Google Sheets) and redo it in the other app, or explore the Interactive Cheat Sheet's SQL column.

Lesson 1: Spreadsheet Foundations

Workspace: **Excel 365** · 10 challenges · about 50 minutes

Spreadsheets are where most data work starts — including the data used to train and test AI. Modern Excel can answer questions about thousands of rows with one formula.

Students will be able to

- SUM, COUNTIF, SUMIF
- XLOOKUP
- FILTER, SORT, UNIQUE (dynamic arrays)
- IFS and LET
- Absolute vs relative references

Common misconceptions

- #SPILL! means something is in the way of a spilling result — clear the cells below.
- Fill formulas down with Ctrl+D; check the \$ signs before you do.
- A typed number is frozen; a formula updates when the data changes.

Vocabulary

Formula	Starts with = and calculates something.	=SUM(F2:F25)
Range	A rectangle of cells, written first:last.	Products!D2:D25
Relative vs absolute reference	A2 shifts when copied; \$A\$2 stays put.	=E2*\$H\$1
Function argument	The values inside the parentheses, separated by commas.	SUMIF(range, criteria, sum_range)
Dynamic array / spill	One formula returns many results that flow into the cells below.	=FILTER(B2:B25, C2:C25="Drinks")
Lookup	Find a row by a key and return another column from it.	=XLOOKUP("SKU-404", A2:A25, B2:B25)

Challenges & answer key

#	Challenge	Task	Answer
1	Your First Formula ☆☆☆ · 10 XP	In Scratch!B2, find the total number of items in stock (the InStock column on the Products sheet). Takeaway: SheetName!Range lets one sheet use data from another sheet.	Scratch!B2 =SUM(Products!F2:F25)

2	What Is It All Worth? ☆☆☆ · 15 XP	In Scratch!B3, calculate the total value of the inventory: every product's Price × InStock, all added up — in ONE formula. Takeaway: SUMPRODUCT works on every version. SUM(D2:D25*F2:F25) only works in Excel 365 (or with Ctrl+Shift+Enter in old Excel).	Scratch!B3 <code>=SUMPRODUCT(Products!D2:D25,Products!F2:F25)</code> Also accepted: <code>=SUM(Products!D2:D25*Products!F2:F25)</code>
3	Count the Categories ☆☆☆ · 15 XP	In Scratch!B4, count how many products are in the School Supplies category. Takeaway: COUNTIF and SUMIF work the same in every spreadsheet app.	Scratch!B4 <code>=COUNTIF(Products!C2:C25,"School Supplies")</code>
4	Store S03 Sales ☆☆☆ · 20 XP	In Scratch!B5, add up the Qty of every sale made by store S03 (Sales sheet). Takeaway: SUMIF = "add these numbers, but only on rows where something is true." In SQL it is SUM(...) WHERE ...	Scratch!B5 <code>=SUMIF(Sales!C2:C61,"S03",Sales!E2:E61)</code>
5	XLOOKUP Detective ☆☆☆ · 20 XP	Type SKU-404 into Scratch!A7. In Scratch!B7, use XLOOKUP to show that SKU's product name. It should still work if someone types a different SKU into A7! Takeaway: XLOOKUP only exists in Excel 365/2021 and Google Sheets. Excel 2013 shows #NAME? — try it there next!	Scratch!A7 SKU-404 Scratch!B7 <code>=XLOOKUP(A7,Products!A2:A25,Products!B2:B25)</code>
6	Spill the Drinks ☆☆☆ · 25 XP	In Scratch!D2, write ONE formula that lists the names of every product in the Drinks category. The answers should spill down. Takeaway: If anything is typed in the cells below D2, Excel shows #SPILL! because the answer has no room.	Scratch!D2 <code>=FILTER(Products!B2:B25,Products!C2:C25="Drinks")</code>
7	Sorted & Unique ☆☆☆ · 25 XP	In Scratch!F2, list each product category once, sorted A→Z, in one formula. Takeaway: Nesting = putting a function inside another function. The inside one runs first.	Scratch!F2 <code>=SORT(UNIQUE(Products!C2:C25))</code>
8	Stock Status with IFS ☆☆☆ · 30 XP	On the Products sheet, type Status in I1. In I2:I25 show "Reorder" if InStock is below ReorderAt, "Low" if InStock is below ReorderAt × 1.5, otherwise "OK". Use IFS. Takeaway: IFS is new. In Excel 2013 you would need IF inside IF inside IF.	Products!I1 Status Products!I2:I25 (type in the first cell, fill down) <code>=IFS(F2<G2,"Reorder",F2<G2*1.5,"Low",TRUE,"OK")</code>
9	LET It Be ☆☆☆ · 30 XP	In Scratch!B9, calculate the profit margin of SKU-502: (price – cost) ÷ price. Use LET to name the price and the cost.	Scratch!B9 <code>=LET(p,XLOOKUP("SKU-502",Products!A2:A25,Products!D2:</code>

Takeaway: LET makes long formulas readable — and Excel only calculates each name once.

D25), c, XL00KUP("SKU-502", Products!A2:A25, Products!E2:E25), (p - c) / p)

10 Build a Revenue Column

★★★ · 35 XP

On the Sales sheet, type Revenue in G1. Fill G2:G61 with Qty × Price. The price is on the Products sheet, so you will need a lookup.

Takeaway: \$A\$2 is an ABSOLUTE reference: it does not change when copied. A2 is RELATIVE: it shifts.

Sales!G1

Revenue

Sales!G2:G61 (type in the first cell, fill down)

=E2*XL00KUP(D2, Products!\$A\$2:\$A\$25, Products!\$D\$2:\$D\$25)

Discussion

1. Why is a formula better than typing the answer, even if the answer is right today?
2. When would you use \$A\$2 instead of A2?

Exit ticket

Write a formula that counts how many products cost more than \$10.

Answer: =COUNTIF(Products!D2:D25, ">10")

Lesson 2: Classic Excel & Nested Functions

Workspace: **Excel 2013** · 8 challenges · about 50 minutes

Not everyone has the newest software. Building answers from classic functions teaches you how lookups and logic really work — and your files will open on any computer.

Students will be able to

- INDEX + MATCH
- VLOOKUP exact match
- Nested IF
- IFERROR
- Ctrl+Shift+Enter array formulas

Common misconceptions

- Forgetting FALSE in VLOOKUP gives wrong answers with no error.
- Without Ctrl+Shift+Enter, array formulas quietly use only one row.
- Count your parentheses: every IF(needs a).

Vocabulary

Nesting	Putting one function inside another. The inside one runs first.	<code>=IF(A2>10, "Big", IF(A2>5, "Medium", "Small"))</code>
INDEX / MATCH	MATCH finds the position; INDEX returns the value at that position.	<code>=INDEX(B2:B25, MATCH("SKU-404", A2:A25, 0))</code>
Exact vs approximate match	VLOOKUP needs FALSE for exact matches.	<code>=VLOOKUP(x, A2:D25, 3, FALSE)</code>
Array formula (CSE)	A formula that works on whole ranges, confirmed with Ctrl+Shift+Enter.	<code>{=MAX(IF(C2:C25="Snacks", D2:D25))}</code>
#NAME?	The program does not know a function name — often because it is too new.	<code>=XL00KUP(...)</code> in Excel 2013

Challenges & answer key

#	Challenge	Task	Answer
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1	Back to 2013 ★★★ · 10 XP	In Scratch!B2, try a modern formula: =XLOOKUP("SKU-101", Products!A2:A25, Products!B2:B25). What does Excel 2013 do? Takeaway: #NAME? means "I don't know that word." Old Excel has never heard of XLOOKUP.	Scratch!B2 =XLOOKUP("SKU-101", Products!A2:A25, Products!B2:B25) The correct result is #NAME? — the point is to see it.
2	INDEX + MATCH ★★★ · 25 XP	In Scratch!B3, show the product name for SKU-404 using INDEX and MATCH nested together (no VLOOKUP). Takeaway: INDEX/MATCH works in EVERY version of Excel and Google Sheets. It is the universal lookup.	Scratch!B3 =INDEX(Products!B2:B25, MATCH("SKU-404", Products!A2:A25, 0))
3	VLOOKUP's Trap ★★★ · 20 XP	In Scratch!B4, use VLOOKUP to find the price of the Hoodie by its name. Make sure you get the RIGHT price! Takeaway: Always finish VLOOKUP with FALSE (exact match). Forgetting it is the #1 VLOOKUP bug in the world.	Scratch!B4 =VLOOKUP("Hoodie", Products!B2:D25, 3, FALSE) Without FALSE the answer is \$5.99 (wrong) — a great discussion point.
4	The Nested IF Ladder ★★★ · 30 XP	On Products, fill I2:I25 with "Reorder" (InStock below ReorderAt), "Low" (below ReorderAt × 1.5) or "OK". IFS doesn't exist here — nest IF inside IF. Takeaway: Every IFS can be rewritten as nested IFs. Count your parentheses — each IF(needs a).	Products!I2:I25 (type in the first cell, fill down) =IF(F2<G2, "Reorder", IF(F2<G2*1.5, "Low", "OK"))
5	IFERROR Safety Net ★★★ · 20 XP	Type any SKU in Scratch!A6. In Scratch!B6, show its product name — or the words Not found if the SKU doesn't exist. Takeaway: IFERROR catches any error. IFNA (Excel 2013+) only catches #N/A, which is safer for lookups.	Scratch!A6 SKU-101 Scratch!B6 =IFERROR(VLOOKUP(A6, Products!A2:B25, 2, FALSE), "Not found")
6	The Ctrl+Shift+Enter Club ★★★ · 40 XP	In Scratch!B7, find the highest Price in the Electronics category. MAXIFS does not exist in 2013! Use MAX(IF(...)) and confirm with Ctrl+Shift+Enter. Takeaway: Before dynamic arrays, formulas that work on whole ranges needed Ctrl+Shift+Enter. Without it, Excel quietly used only ONE row.	Scratch!B7 =MAX(IF(Products!C2:C25="Electronics", Products!D2:D25)) Ctrl+Shift+Enter Confirm with Ctrl+Shift+Enter. Plain Enter gives a wrong answer.
7	Build a Price Tag ★★★ · 15 XP	In Scratch!B8, build the text Granola Bar - \$1.50 from Products!B2 and Products!D2. Takeaway: Formatting (\$) is not part of the number. TEXT(...) bakes the format into text.	Scratch!B8 =Products!B2&" - "&TEXT(Products!D2, "\$0.00")

8 Wildcard SUMIFS

★★★ · 30 XP

In Scratch!B9, add up the Qty of sales from store S02 for products whose SKU starts with SKU-3 (school supplies).

Takeaway: SUMIFS/COUNTIFS have existed since Excel 2007, so they are safe to use everywhere.

Scratch!B9

```
=SUMIFS(Sales!E2:E61, Sales!C2:C61, "S02", Sales!D2:D61, "SKU-3*")
```

Discussion

1. Your formula works at school but shows #NAME? at home. What are two possible reasons?
2. Why did people love SUMPRODUCT before dynamic arrays existed?

Exit ticket

Rewrite =XLOOKUP("SKU-201", A2:A25, B2:B25) so it works in Excel 2013.

Answer: `=INDEX(B2:B25, MATCH("SKU-201", A2:A25, 0))`

Lesson 3: Google Sheets Survival Guide

Workspace: **Google Sheets** · 9 challenges · about 50 minutes

Google Sheets runs on every Chromebook and is built for sharing. It looks like Excel, but its functions do not always take the same arguments — the #1 reason "it worked at school but not at home".

Students will be able to

- FILTER and SORT the Sheets way
- ARRAYFORMULA
- QUERY
- SPLIT and REGEXMATCH
- Open-ended ranges

Common misconceptions

- SORT(..., -1) sorts ASCENDING in Sheets — no error, wrong answer.
- CONCAT only joins two values in Sheets.
- A bad formula is saved and shows #ERROR! instead of a pop-up.

Vocabulary

ARRAYFORMULA	Lets normal formulas work on whole ranges and fill a column with one formula.	<code>=ARRAYFORMULA(D2:D25-E2:E25)</code>
FILTER conditions	Each condition is its own argument (AND).	<code>=FILTER(B2:B25, C2:C25="Snacks", D2:D25>1)</code>
is_ascending	SORT direction is TRUE/FALSE in Sheets.	<code>=SORT(B2:D25, 3, FALSE)</code>
QUERY	SQL-like questions inside a spreadsheet; columns are letters.	<code>=QUERY(A1:H25, "select C, sum(F) group by C")</code>
Open-ended range	E2:E means "E2 to the bottom".	<code>=SUM(Sales!E2:E)</code>

Challenges & answer key

#	Challenge	Task	Answer
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1	Excel Habits, Sheets Rules ★★★ · 20 XP	In Scratch!B2, type the Excel-style formula <code>=FILTER(Products!B2:B25, Products!C2:C25="Apparel", "None")</code>. Read the error, then fix it so it lists the Apparel products. Takeaway: Same function name, different arguments. In Sheets every extra FILTER argument is another condition.	Scratch!B2 <code>=FILTER(Products!B2:B25, Products!C2:C25="Apparel")</code>
2	Two Conditions ★★★ · 25 XP	In Scratch!D2, list Snacks that cost more than \$1.00. Give FILTER the two conditions as two separate arguments (the Sheets way). Takeaway: Excel needs one combined test instead: <code>(cond1)*(cond2)</code>. The translator panel shows both!	Scratch!D2 <code>=FILTER(Products!B2:B25, Products!C2:C25="Snacks", Products!D2:D25>1)</code>
3	Which Way Is Down? ★★★ · 20 XP	In Scratch!F2, sort Products!B2:D25 by price, highest first. (Hint: Excel's -1 will NOT work here — try it and see!) Takeaway: In Sheets, -1 counts as TRUE (any non-zero number is TRUE) so it sorted the WRONG way with no error. Silent bugs are the worst bugs.	Scratch!F2 <code>=SORT(Products!B2:D25, 3, FALSE)</code>
4	One Formula, 24 Answers ★★★ · 30 XP	On Products, put ONE formula in I2 that fills I2:I25 with the profit per unit (Price – Cost). Use ARRAYFORMULA. Takeaway: In Excel 365 this just works without ARRAYFORMULA. In Excel 2013 it needs Ctrl+Shift+Enter across the whole range.	Products!I2 <code>=ARRAYFORMULA(D2:D25-E2:E25)</code>
5	QUERY: SQL Inside a Spreadsheet ★★★ · 35 XP	In Scratch!J2, use QUERY to show each category and its total InStock. Takeaway: QUERY is Google-only. It is basically SQL — head over to SQL mode to see the real thing.	Scratch!J2 <code>=QUERY(Products!A1:H25, "select C, sum(F) group by C")</code>
6	SPLIT the Manager ★★★ · 15 XP	In Scratch!B10, split the manager name in Stores!E2 into first name (B10) and last name (C10). Takeaway: SPLIT is Google-only. Excel uses Data ► Text to Columns (or TEXTSPLIT in newer Excel 365).	Scratch!B10 <code>=SPLIT(Stores!E2, " ")</code>
7	Ranges With No Bottom ★★★ · 20 XP	In Scratch!B12, add up all Qty on the Sales sheet using an open-ended range that starts at row 2 and has no last row. Takeaway: Open ranges grow automatically as you add rows. In Excel you would use E:E	Scratch!B12 <code>=SUM(Sales!E2:E)</code>

or a Table.

8 The CONCAT Gotcha ★☆☆ · 15 XP	In Scratch!B13, build Cumming, GA from Stores!B2 and Stores!C2. (Try =CONCAT(Stores!B2, ", ", Stores!C2) first and read the error!) Takeaway: The same function name can have different rules in different apps.	Scratch!B13 =CONCATENATE(Stores!B2, ", ", Stores!C2) Also accepted: =Stores!B2&"&Stores!C2
9 Pattern Hunter ★★★ · 25 XP	In Scratch!B14, count how many product names contain a digit (0-9) using REGEXMATCH. Takeaway: Regular expressions are a mini-language for patterns. Google Sheets has them built in.	Scratch!B14 =SUMPRODUCT(- - REGEXMATCH(Products!B2:B25, "\d"))

Discussion

1. Which is worse: a formula that errors, or one that silently gives the wrong answer? Why?
2. Name one thing Google Sheets can do that Excel cannot, and one the other way around.

Exit ticket

In Google Sheets, sort Products!B2:D25 by price, lowest first.

Answer: =SORT(Products!B2:D25, 3, TRUE)

Lesson 4: Data Files: CSV & TSV

Workspace: **CSV / TSV** · 8 challenges · about 45 minutes

CSV is how data moves between programs — and how most AI training data is shared. It is plain text: if you understand the rules, you can read and fix any dataset.

Students will be able to

- Values vs formulas
- Quoting and escaping
- Delimiters
- How spreadsheets change data on import

Common misconceptions

- Saving as CSV throws away formulas, formatting and other sheets.
- Opening a CSV in a spreadsheet can drop leading zeros and mangle long IDs.
- An extra comma in one row shifts every value after it.

Vocabulary

Delimiter	The character between values: comma (CSV) or tab (TSV).	SKU-101,Granola Bar,Snacks
Field / record	A field is one value; a record is one line (row).	4 fields: a,b,c,d
Quoting	Values containing the delimiter, quotes or line breaks are wrapped in "double quotes".	"Gel Pens, 4-pack"
Escaping	A quote inside a quoted value is written twice.	"12"" Ruler"
Type inference	When a program opens a CSV it GUESSES types — and can change your data.	05401 → 5401

Challenges & answer key

#	Challenge	Task	Answer
1	Where Did My Formula Go? ☆☆☆ · 15 XP	You save an Excel workbook as CSV. Cell B2 had =SUM(D2:D25) showing \$189.63. What is stored in the CSV file? Takeaway: CSV keeps the RESULT, not the formula. Re-open it and your formulas are gone. Try "Export from a spreadsheet" to see it happen.	Answer: 189.63
2	Export a Sheet ☆☆☆ · 15 XP	Click Export from spreadsheet, choose the Excel 365 → Products sheet and	Export from spreadsheet ► Excel 365 ► Products ► CSV.

export it as CSV. Look at the "What was lost" report.

Takeaway: Any formulas you added on that sheet became plain numbers in the CSV.

3	Fix the Broken Comma ★★★ · 25 XP	Open broken_products.csv. One row has too many fields because a product name contains a comma. Fix it so every row has 8 fields. Takeaway: Quotes protect commas inside values. This is why TSV (tabs) is popular — names rarely contain tabs.	In broken_products.csv change Gel Pens, 4-pack to "Gel Pens, 4-pack"
4	Commas to Tabs ★★★ · 20 XP	Create a new file named stores.tsv that holds the same data as stores.csv, but separated by TAB characters. Takeaway: Same data, different delimiter. Spreadsheets can open both.	New file stores.tsv, paste stores.csv, Find & replace , → \t (Replace all).
5	The Vanishing Zero ★★★ · 15 XP	Open suppliers.csv and switch the preview to "Open like a spreadsheet would". What happens to Lakeside Apparel's ZIP code 05401? Takeaway: ZIP codes, phone numbers and IDs are TEXT, not numbers. Import them as text or the zeros disappear forever.	Answer: It becomes 5401
6	Quotes Inside Quotes ★★★ · 25 XP	Add a new product to the end of products.csv: SKU SKU-307, product 12" Ruler (that's 12 inches!), category School Supplies, price 1.99, cost 0.6, InStock 30, ReorderAt 10, supplier Office Hub Supply. Takeaway: CSV escaping rule: wrap in quotes, double any inner quotes.	Add to products.csv: SKU-307,"12"" Ruler",School Supplies,1.99,0.6,30,10,Office Hub Supply
7	Count the Fields ★★★ · 10 XP	How many fields (values) are in this CSV line?SKU-302,"Gel Pens, 4-pack",School Supplies,4.5 Takeaway: SKU-302 Gel Pens, 4-pack School Supplies 4.5 → 4 fields.	Answer: 4
8	Pick the Right Format ★★★ · 10 XP	Which file type keeps your formulas, formatting and multiple sheets? Takeaway: Use .xlsx (or a Google Sheet) for working files. Use CSV/TSV to move data between programs.	Answer: .xlsx

Discussion

1. You email a coworker a CSV of your gradebook. What did you lose?
2. Why would anyone choose TSV over CSV?

Exit ticket

Write the CSV line for: Smith, Jo | 05401 | said "hi"

Answer: `"Smith, Jo",05401,"said ""hi"""`

Lesson 5: Data Wrangling I: Cleaning Data for AI

Workspace: **Data Wrangling Lab** · 11 challenges · about 90 minutes

Data scientists spend most of their time cleaning data. An AI model learns whatever is in its training data — including typos, duplicates and fake zeros. Clean data is the difference between a useful model and a confident wrong one.

Students will be able to

- Leading zeros and text vs numbers
- Finding blank and fake-null values
- Converting text to numbers
- Fixing mixed date formats
- TRIM, PROPER, UPPER, nested text formulas

Common misconceptions

- Replacing missing values with 0 changes averages — missing is not zero.
- 03.09.2026 is September 3 in Europe and March 9 in the US.
- Clean into NEW columns so you can always compare with the original.

Vocabulary

Data type	Number, text, date, TRUE/FALSE. Digits can be TEXT ("0104").	=ISTEXT(C2)
Null / missing value	No value. Can hide as blanks or as words like N/A, -, null.	=COUNTBLANK(E2:E31)
Type conversion	Changing a value's type.	=VALUE("12") =TEXT(104, "0000")
Normalization	Making the same thing look the same: case, spaces, formats.	=UPPER(TRIM(B2))
Date serial	Real dates are numbers (days since 1900), so they sort and subtract.	=DATEVALUE("Sep 4, 2026")
Garbage in, garbage out	A model can only be as good as its data.	Open the 🗑️ AI check tab

Challenges & answer key

#	Challenge	Task	Answer
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1	Where Did the Zero Go? ★★☆☆ · 10 XP	RawOrders!C2 shows 104, but the real item code is 0104. Why? Takeaway: IDs, ZIP codes and phone numbers are TEXT, even though they are made of digits. An AI model will treat 104 and "0104" as two different items.	Answer: It was stored as a NUMBER, and numbers never keep leading zeros
2	Rebuild the Item Codes ★★☆☆ · 20 XP	Type CleanCode in RawOrders!I1. In I2:I31, turn every ItemCode (column C) into a 4-character text code like 0104 — whether it arrived as 104, "0104" or "0402". Takeaway: TEXT(x, "0000") is the spreadsheet version of zero-padding. SQL: printf('%04d', x). Python: str(x).zfill(4).	RawOrders!I1 CleanCode RawOrders!I2:I31 (type in the first cell, fill down) =TEXT(C2, "0000")
3	Count the Holes ★★☆☆ · 10 XP	In Report!B2, count how many Qty cells (RawOrders!E2:E31) are completely empty. Takeaway: Empty is only ONE kind of missing data. Look at the next challenge...	Report!B2 =COUNTBLANK(RawOrders!E2:E31)
4	Fake Nulls ★★☆☆ · 20 XP	Some systems write missing values as words. In Report!B3, count Qty cells that say N/A, - or null. Takeaway: COUNTBLANK said 2 — the real number of missing quantities is 5. "null", "N/A" and "-" are TEXT that MEANS missing. Before training a model, they must all become one true blank/NA.	Report!B3 =COUNTIF(RawOrders!E2:E31, "N/A") +COUNTIF(RawOrders!E2:E31, "-") +COUNTIF(RawOrders!E2:E31, "null")
5	One Kind of Missing ★★☆☆ · 25 XP	Type CleanQty in J1. In J2:J31, turn Qty into a real number — "12" (text) becomes 12 — and turn every kind of missing value into a blank "". Takeaway: Text that looks like a number does not add up: SUM ignores "12". Converting types is step one of every data-cleaning job.	RawOrders!J1 CleanQty RawOrders!J2:J31 (type in the first cell, fill down) =IFERROR(VALUE(E2), "")
6	Same Store, Different Spelling ★★☆☆ · 15 XP	Type CleanStore in K1. In K2:K31, clean the Store codes so "s04" and " S02 " become S04 and S02. Takeaway: To a computer "S04", "s04" and " S04 " are three different stores. A model trained on this would think you have 13 stores!	RawOrders!K1 CleanStore RawOrders!K2:K31 (type in the first cell, fill down) =UPPER(TRIM(B2))
7	Prices That Are Really Text ★★☆☆ · 15 XP	Type CleanPrice in L1. In L2:L31, turn UnitPrice into real numbers (" \$3.25 " → 3.25, "14.99 " → 14.99).	RawOrders!L1 CleanPrice RawOrders!L2:L31 (type in the first cell, fill down)

	Takeaway: If a column has even ONE text price, averages and sums silently skip it. Always check types.	<code>=VALUE (F2)</code>
8 Four Kinds of Dates ★★★ · 30 XP	Type CleanDate in M1. In M2:M31, make every OrderDate a real date. Some are already dates, others are text like "2026-09-12", "9/23/2026" or "Sep 4, 2026". (Row 19 is special — skip it for now.) Takeaway: Real dates are numbers (days since 1900), so you can sort them, subtract them and group them by month. Text dates cannot do any of that.	RawOrders!M1 <code>CleanDate</code> RawOrders!M2:M31 (type in the first cell, fill down) <code>=IF (ISNUMBER (D2) , D2 , DATEVALUE (D2)))</code> Row 19 (European date) shows #VALUE! here — that is the next challenge.
9 The European Date ★★★ · 30 XP	RawOrders!D19 says 03.09.2026 — in Europe that means 3 September 2026. DATEVALUE cannot read it. In Report!B5, build the real date from its pieces. Takeaway: Is 03/09 March 9th or September 3rd? It depends on the country. Merging data from different places is one of the most common ways datasets go wrong.	Report!B5 <code>=DATE (RIGHT (RawOrders!D19,4) , MID (RawOrders!D19,4,2) , LEFT (RawOrders!D19,2))</code>
10 Fix the Names ★★★ · 35 XP	Type CleanName in N1. In N2:N31, turn every Customer into First Last with proper capitals: "jordan smith", " Avery Johnson ", "TAYLOR KIM" and even "Davis, Morgan" → "Morgan Davis". Takeaway: This is a nested formula: IF, ISNUMBER, FIND, PROPER, TRIM, MID and LEFT working together. Building it one piece at a time is the skill.	RawOrders!N1 <code>CleanName</code> RawOrders!N2:N31 (type in the first cell, fill down) <code>=IF (ISNUMBER (FIND (" , " , G2)) , PROPER (TRIM (MID (G2 , FIND (" , " , G2) +1 , 99))) & " " & LEFT (G2 , FIND (" , " , G2) - 1) , PROPER (TRIM (G2))))</code>
11 How Dirty Was It? ★★★ · 10 XP	After CleanQty is done, count in Report!B4 how many orders have no usable quantity. Takeaway: Report your missing data. A model trained without telling anyone that 1 in 6 rows was dropped can be badly biased.	Report!B4 <code>=COUNTBLANK (RawOrders!J2:J31)</code> Needs earlier step(s): wr-qty

Discussion

1. An AI model was trained on data where missing quantities were replaced with 0. What will it get wrong?
2. Is 03/04/2026 March 4 or April 3? How would you find out for a real dataset?
3. Why clean into a NEW column instead of overwriting the original?

Exit ticket

A column of ZIP codes shows 2110 and 5401. Write the formula that fixes them.

Answer: `=TEXT (A2, "00000")`

Lesson 6: Data Wrangling II: Combining & Lookups

Workspace: **Data Wrangling Lab** · 11 challenges · about 90 minutes

Real datasets live in separate tables. Joining them — matching orders to products, stores to regions — only works when the keys match exactly. This is the same idea as a JOIN in SQL or merge() in Python.

Students will be able to

- VLOOKUP and HLOOKUP
- Why lookups fail (type mismatch)
- Concatenation and delimiters
- TEXTJOIN
- Filtering around missing values

Common misconceptions

- Clean the key column BEFORE looking up.
- Lock lookup ranges with \$ before filling down.
- Blank text "" counts as bigger than any number in comparisons.

Vocabulary

Key	The column used to match rows between tables.	Item code 0104
VLOOKUP / HLOOKUP	Look up down the first column / across the first row.	=HLOOKUP("S03", Targets!A1:F3, 3, FALSE)
Type mismatch	Number 104 never equals text "0104" — the lookup returns #N/A.	=VLOOKUP(C2, ...) → #N/A
Concatenation	Joining text together.	=B2 & ", " & A2
Delimited list	Several values in one cell separated by a character.	snacks drinks apparel
Filtering	Keeping only rows that pass a test — watch out for blanks.	=FILTER(A2:A31, ISNUMBER(J2:J31)*(J2:J31>10))

Challenges & answer key

#	Challenge	Task	Answer
1	The #N/A Mystery ☆☆☆ · 15 XP	=VLOOKUP(RawOrders!C2, ItemCodes!A2:D25, 2, FALSE) gives #N/A, even though code 0104 IS in the ItemCodes table. Why? Takeaway: A lookup only matches the same TYPE: number 104 ≠ text "0104". This is the	Answer: C2 is the NUMBER 104 and the table has the TEXT "0104" — they never match

#1 reason lookups fail on real data.

2	Look Up the Products ★☆☆ · 20 XP	Type Product in O1. In O2:O31, use VLOOKUP with your CleanCode (column I) to show each product name from ItemCodes. Takeaway: Clean first, then join. In SQL this is a JOIN; in Python pandas it is merge().	RawOrders!O1 Product RawOrders!O2:O31 (type in the first cell, fill down) <code>=VLOOKUP(I2,ItemCodes!\$A\$2:\$D\$25,2,FALSE)</code> Needs earlier step(s): wr-code
3	Line Totals ★☆☆ · 25 XP	Type LineTotal in P1. In P2:P31, calculate CleanQty × the ItemCodes price. If CleanQty is blank, leave LineTotal blank. Takeaway: Missing values spread: blank × price should stay blank, not turn into 0 — a fake 0 would teach a model that people order nothing.	RawOrders!P1 LineTotal RawOrders!P2:P31 (type in the first cell, fill down) <code>=IF(J2="", "", J2*XLOOKUP(I2,ItemCodes!\$A\$2:\$A\$25,ItemCodes!\$D\$2:\$D\$25))</code> Needs earlier step(s): wr-code, wr-qty
4	HLOOKUP Goes Sideways ★☆☆ · 15 XP	The Targets sheet is laid out sideways (stores across the top). In Report!B7, use HLOOKUP to find the Sept Target for store S03. Takeaway: VLOOKUP = vertical tables, HLOOKUP = horizontal tables. XLOOKUP and INDEX/MATCH handle both.	Report!B7 <code>=HLOOKUP("S03",Targets!A1:F3,3,FALSE)</code>
5	Region for Every Order ★☆☆ · 20 XP	Type Region in Q1. In Q2:Q31, use HLOOKUP with your CleanStore (column K) to show each order's region. Takeaway: Every lookup in this lab worked only because you cleaned the key column first.	RawOrders!Q1 Region RawOrders!Q2:Q31 (type in the first cell, fill down) <code>=HLOOKUP(K2,Targets!\$B\$1:\$F\$2,2,FALSE)</code> Needs earlier step(s): wr-store
6	Now the Math Works ★☆☆ · 10 XP	In Report!B8, add up all the cleaned quantities (CleanQty). Takeaway: SUM(E2:E31) gives a smaller, WRONG total because it skips text numbers. Clean data changes the answer.	Report!B8 <code>=SUM(RawOrders!J2:J31)</code> Needs earlier step(s): wr-qty
7	Last, First ★☆☆ · 15 XP	On Contacts, fill G2:G9 with names written as Last, First (e.g. "Whitfield, Dana"). Takeaway: A delimiter is the character that separates pieces of text — here it is ", ".	Contacts!G2:G9 (type in the first cell, fill down) <code>=B2&"", "&A2</code>
8	Keep the Zero (Again) ★☆☆ · 25 XP	In Contacts!H2:H9, build a mailing line like Burlington, VT 05401. Watch out: the Zip column lost its leading zeros! Takeaway: ZIP codes in New England start with 0. Treat ZIPs as text from the start and	Contacts!H2:H9 (type in the first cell, fill down) <code>=C2&"", "&D2&" "&TEXT(E2,"00000")</code>

this problem never happens.

9	Count the Delimited Tags ★★★ · 30 XP	Column F holds tags separated by a pipe: snacks drinks. In Contacts!I2:I9, count how many tags each contact has (blank = 0). Takeaway: Sheets has SPLIT(F2, " ") and Excel 365 has TEXTSPLIT to break the tags into cells. Counting delimiters is the trick that works everywhere.	Contacts!I2:I9 (type in the first cell, fill down) =IF (F2="",0,LEN(F2) - LEN(SUBSTITUTE(F2," ",""))+1)
10	Write a CSV Line ★★★ · 15 XP	In Report!B10, join Contacts!A2:E2 into one CSV line with commas between the values. Takeaway: This is exactly what "Save as CSV" does for every row. Now you know why a comma INSIDE a value breaks the file.	Report!B10 =TEXTJOIN(", ",FALSE,Contacts!A2:E2)
11	Filter the Feed ★★★ · 30 XP	In Report!D2, use FILTER to list the OrderIDs whose CleanQty is more than 10. (Careful — try J2:J31>10 alone first and look at what sneaks in!) Takeaway: Missing values break filters in sneaky ways. Always ask "what happens to the blanks?"	Report!D2 =FILTER(RawOrders!A2:A31,ISNUMBER(RawOrders!J2:J31)*(RawOrders!J2:J31>10)) Needs earlier step(s): wr-qty

Discussion

1. VLOOKUP says #N/A but you can SEE the value in the table. List three reasons this happens.
2. Why do joins (lookups) come AFTER cleaning, never before?

Exit ticket

Count the items in "red|green|blue" with one formula that works in every app.

Answer: =LEN(A2) - LEN(SUBSTITUTE(A2,"|",""))+1

Lesson 7: SQL for Data Analysts

Workspace: **SQL Database** · 9 challenges · about 75 minutes

Companies keep their data in databases, and SQL is how you ask them questions. Every spreadsheet skill has a SQL twin — and SQL scales to millions of rows.

Students will be able to

- SELECT, WHERE, ORDER BY
- GROUP BY and HAVING
- JOIN (the SQL VLOOKUP)
- Integer division
- UPDATE safely

Common misconceptions

- Text uses 'single quotes'.
- Whole-number division drops decimals: $7 / 2 = 3$.
- UPDATE or DELETE without WHERE changes every row.

Vocabulary

SELECT ... FROM	Choose columns from a table.	<code>SELECT product, price FROM products;</code>
WHERE	Keep only matching rows (like FILTER).	<code>WHERE category = 'Snacks'</code>
GROUP BY	One result per group (like SUMIF for every value at once).	<code>SELECT store_id, SUM(qty) FROM sales GROUP BY store_id;</code>
JOIN	Combine tables on a key (like VLOOKUP for every row).	<code>FROM sales s JOIN products p ON s.sku = p.sku</code>
NULL	SQL's missing value. Test with IS NULL, never = NULL.	<code>WHERE qty IS NULL</code>

Challenges & answer key

#	Challenge	Task	Answer
1	SELECT Everything ☆☆☆ · 10 XP	Show every column and every row of the products table. <i>Takeaway: A spreadsheet shows all data all the time. A database only shows what you ask for.</i>	<code>SELECT * FROM products;</code>
2	WHERE = FILTER ☆☆☆ · 15 XP	Show the product and price of every product in the Electronics category.	<code>SELECT product, price FROM products WHERE category =</code>

		Takeaway: WHERE works like FILTER. Remember: 'single quotes' for text, and it is case-sensitive.	'Electronics';
3	Top 3 Most Expensive ★★★ · 20 XP	Show the product and price of the 3 most expensive products, highest first. Takeaway: ORDER BY = SORT. LIMIT = "only the first N".	<pre>SELECT product, price FROM products ORDER BY price DESC LIMIT 3;</pre>
4	GROUP BY = SUMIF for Everyone ★★★ · 25 XP	Show each store_id with the total qty it sold (from the sales table). Takeaway: One GROUP BY query does what five SUMIF formulas would.	<pre>SELECT store_id, SUM(qty) FROM sales GROUP BY store_id;</pre>
5	JOIN = VLOOKUP ★★★ · 30 XP	For every sale, show the order_id and the product name. The name lives in the products table, so you need to JOIN. Takeaway: A JOIN does a lookup for EVERY row at once — no filling down.	<pre>SELECT s.order_id, p.product FROM sales s JOIN products p ON s.sku = p.sku;</pre>
6	HAVING: Filter the Groups ★★★ · 30 XP	Show the categories that have more than 5 products, with their count. Takeaway: WHERE → rows. HAVING → groups.	<pre>SELECT category, COUNT(*) FROM products GROUP BY category HAVING COUNT(*) > 5;</pre>
7	Revenue by City ★★★ · 40 XP	Show each store's city and its total revenue (qty × price), highest first. You need sales, stores AND products. Takeaway: This would take lookups, a helper column and a pivot table in a spreadsheet. In SQL it is one query.	<pre>SELECT st.city, SUM(s.qty * p.price) AS revenue FROM sales s JOIN stores st ON s.store_id = st.store_id JOIN products p ON s.sku = p.sku GROUP BY st.city ORDER BY revenue DESC;</pre>
8	The Integer Division Trap ★★★ · 20 XP	Find the average qty per sale with its decimals (it is NOT a whole number). Try SELECT SUM(qty) / COUNT(*) FROM sales; first and see what goes wrong. Takeaway: In SQL, 7 / 2 = 3. In a spreadsheet, 7 / 2 = 3.5. Same math, different rules!	<pre>SELECT AVG(qty) FROM sales;</pre> SUM(qty)/COUNT(*) gives 4 (integer division) — the trap.
9	Restock Day ★★★ · 25 XP	The truck arrived! For every product where in_stock is below reorder_at, set in_stock to reorder_at × 2. Leave the others alone. (If you mess up, use Reset database.) Takeaway: Forget the WHERE and EVERY product changes. Real databases have no Ctrl+Z!	<pre>UPDATE products SET in_stock = reorder_at * 2 WHERE in_stock < reorder_at;</pre>

Discussion

1. Which spreadsheet function is most like GROUP BY? Like JOIN? Like WHERE?

2. Why do real databases not have an Undo button — and what do companies do instead?

Exit ticket

Write SQL for the number of sales per store.

Answer: `SELECT store_id, COUNT(*) FROM sales GROUP BY store_id;`

Lesson 8: Cross-Platform Translator

Workspace: **Compare** · 5 challenges · about 25 minutes

The same formula can work, fail, or silently give a different answer depending on the app. Checking before you share is a professional habit.

Students will be able to

- Which functions exist where
- Silent differences between apps
- Spreadsheet ideas in SQL

Common misconceptions

- Test your file in the app your audience will use.
- Prefer functions that exist everywhere when sharing widely.

Vocabulary

Compatibility	Whether a formula works in another app or version.	XLOOKUP: Excel 365 ✓, Excel 2013 x
Silent difference	No error, but a different result — the most dangerous kind.	SORT(..., -1) in Sheets
Translation	Rewriting a formula so it works elsewhere.	XLOOKUP → INDEX/MATCH

Challenges & answer key

#	Challenge	Task	Answer
1	Who Says #NAME? ☆☆☆ · 10 XP	Where does =XLOOKUP(A2, B:B, C:C) give a #NAME? error? <i>Takeaway: Excel 2013 never got XLOOKUP. INDEX/MATCH works everywhere.</i>	Answer: Excel 2013
2	The Silent Sort ☆☆☆ · 15 XP	You type =SORT(A2:C20, 3, -1) in Google Sheets. What happens? <i>Takeaway: No error, wrong answer. Always check your results!</i>	Answer: It sorts low → high
3	Pointing at Nothing ☆☆☆ · 10 XP	Cell A1 is empty. What does =A1 show in Excel vs Google Sheets? <i>Takeaway: Little differences like this break checks such as =IF(B2="", ...).</i>	Answer: Excel: 0, Sheets: blank
4	Array Formulas Through Time	Which one needs Ctrl+Shift+Enter (or ARRAYFORMULA) to make	Answer: Excel 2013 and Google Sheets

★★☆ · 15 XP

=MAX(IF(C2:C25="Snacks", D2:D25))

give the right answer?

Takeaway: Excel 2013 needs

Ctrl+Shift+Enter; Google Sheets needs

ARRAYFORMULA (Ctrl+Shift+Enter adds it for you).

5 Spreadsheet → SQL

★★☆ · 10 XP

Which SQL clause does the same job as VLOOKUP?

Answer: JOIN

Takeaway: JOIN matches rows from two

tables, like a lookup on every row at once.

Discussion

1. Before sharing a spreadsheet, what should you check?

Exit ticket

Name a function that exists in Excel 365 but gives #NAME? in Excel 2013.

Answer: XLOOKUP, FILTER, SORT, UNIQUE, IFS, TEXTJOIN, LET, MAXIFS...

Appendix: what is different in each tool

Excel 365

- Dynamic arrays: one formula can **spill** results into the cells below it. If something is in the way you get `#SPILL!`.
- Has XLOOKUP, XMATCH, FILTER, SORT, UNIQUE, SEQUENCE, LET, IFS, SWITCH, TEXTJOIN, CONCAT, MAXIFS and MINIFS.
- Refer to a whole spilled result with `A2#` (the spill operator). Google Sheets does not have this.
- `SORT(range, 3, -1)` sorts column 3 high→low. Sort order is **1 or -1**.
- `FILTER(range, include, [if_empty])` takes ONE include test. Combine tests with `*` (AND) or `+` (OR).
- Open-ended ranges like `A2:A` are **not** allowed. Use `A:A` or `A2:A1000`.
- A formula pointing at an empty cell shows **0**.
- Excel refuses to accept a formula with a typo and shows a **"There's a problem with this formula"** pop-up.

Excel 2013

- **No** XLOOKUP, FILTER, SORT, UNIQUE, LET, IFS, SWITCH, TEXTJOIN, CONCAT, MAXIFS, MINIFS → typing them gives `#NAME?`.
- Lookups: use `VLOOKUP(..., FALSE)` or the classic combo `INDEX(return_range, MATCH(value, lookup_range, 0))`.
- Instead of IFS, nest IFs: `IF(test1, a, IF(test2, b, c))`.
- No spilling. A formula that works on a whole range must be confirmed with **Ctrl+Shift+Enter**. Excel shows it as `{=MAX(IF(...))}`.
- Without Ctrl+Shift+Enter, a range used where one value is expected is cut down to the value in the **same row** (implicit intersection) — often the wrong answer!
- `SUMPRODUCT` handles arrays without Ctrl+Shift+Enter, which is why it was so popular.
- Join text with `CONCATENATE(a, b, c)` or `a & b & c`.

Google Sheets

- `FILTER(range, condition1, condition2, ...)` — each condition is its **own argument**, and there is **no if_empty**. No matches → `#N/A`.
- `SORT(range, column, is_ascending)` uses **TRUE/FALSE**. Typing Excel's `-1` still sorts ascending (any non-zero number counts as TRUE)!
- Operators on whole ranges (like `D2:D25*F2:F25`) need `ARRAYFORMULA( )`. Ctrl+Shift+Enter adds it for you.
- `CONCAT` only joins **two** values. Use `CONCATENATE`, `JOIN`, `TEXTJOIN` or `&` for more.
- Open-ended ranges like `A2:A` work (from row 2 to the bottom).
- Sheets-only functions: QUERY, ARRAYFORMULA, SPLIT, JOIN, REGEXMATCH, REGEXEXTRACT, REGEXREPLACE, COUNTUNIQUE, IMPORTRANGE, GOOGLEFINANCE.
- A bad formula is still saved — the cell shows `#ERROR!` (parse error) instead of a pop-up.
- A formula pointing at an empty cell shows a **blank**, not 0.

CSV / TSV

- A CSV/TSV file stores **values only**. Saving as CSV turns every formula into its current result.

- No formatting (bold, colors, \$ signs as formats), and only **one sheet** per file.
- A value that contains a comma must be wrapped in "double quotes": "Gel Pens, 4-pack" .
- A double quote inside a quoted value is written twice: "12" " Ruler" .
- TSV uses the TAB character, so commas inside values are no problem.
- Spreadsheets "helpfully" convert values when they open a CSV: 05401 → 5401 , long numbers → 4.41712E+15 .

SQL Database

- Text goes in '**single quotes**'. "Double quotes" mean a column name.
- Text comparisons are **case-sensitive**: 'snacks' does not equal 'Snacks' . LIKE is not case-sensitive.
- Dividing two whole numbers drops the decimals: 7 / 2 = 3. Use 7 / 2.0 or AVG() .
- Columns are referred to by name, not letter: price , not D .
- Joining tables (JOIN) replaces VLOOKUP. Grouping (GROUP BY) replaces SUMIF/COUNTIF.
- UPDATE and DELETE without WHERE change **every row**. There is no undo in a real database!

Appendix: error codes

#NAME?	The formula contains unrecognized text (usually a misspelled or unsupported function name).
#VALUE!	Wrong type of value — e.g. doing math on text, or a range where one value was expected.
#REF!	The formula refers to a cell or sheet that does not exist.
#DIV/0!	Dividing by zero (or by an empty cell).
#N/A	A value is not available — usually a lookup that found no match.
#NUM!	A number is invalid or too big/small.
#NULL!	Ranges do not intersect.
#SPILL!	The result is an array, but cells it needs to spill into are not empty.
#CALC!	The calculation engine hit something it does not support, like an empty array.
#ERROR!	Google Sheets could not understand (parse) the formula.